SF-EDGE Product Requirements Document

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1. Purpose

The Smart Factory Edge Gateway, named SF-EDGE, is an industrial gateway used to collect machine telemetry, normalize it, buffer it locally, and forward it to a cloud analytics platform.

The product is intended for small and medium manufacturing sites where machines use mixed protocols such as Modbus TCP, OPC UA, MQTT, and REST-based sensor adapters.

The gateway must continue operating during temporary network outages and must provide traceable logs for quality and maintenance teams.

2. Business Objectives

The product shall:

- reduce manual collection of machine production data;
- provide near real-time visibility into machine status;
- support predictive maintenance use cases;
- allow remote configuration by authorized users;
- preserve telemetry data during short connectivity outages;
- provide audit trails for configuration and operational events.

3. Users

3.1 Plant Operator

The plant operator monitors machine status and acknowledges operational alerts. This user does not modify protocol mappings or security settings.

3.2 Maintenance Engineer

The maintenance engineer configures device connections, validates signal mappings, and reviews device health.

3.3 Site Administrator

The site administrator manages users, roles, remote access, certificates, and site-level policies.

3.4 Cloud Support Engineer

The cloud support engineer can inspect diagnostic bundles only when the site administrator grants temporary access.

4. Functional Requirements

REQ-FUNC-001 Device Registration

The gateway shall allow a maintenance engineer to register a new industrial device with a unique device ID, protocol type, IP address or hostname, polling interval, and signal map.

Acceptance criteria:

- a registered device appears in the device inventory within 5 seconds;
- duplicate device IDs are rejected;

- invalid IP addresses or hostnames are rejected;
- all registration events are stored in the audit log.

REQ-FUNC-002 Supported Protocols

The gateway shall support the following protocols in version 1:

- Modbus TCP;
- OPC UA;
- MQTT subscriber mode;
- HTTP sensor adapter mode.

Protocol-specific settings shall be validated before the device is saved.

REQ-FUNC-003 Telemetry Collection

The gateway shall collect telemetry according to each device polling interval.

The minimum polling interval is 1 second. The default polling interval is 10 seconds. Polling intervals below 1 second shall be rejected by the API.

REQ-FUNC-004 Local Buffering

If the cloud connection is unavailable, the gateway shall buffer telemetry locally for at least 24 hours at a rate of 2,000 measurements per minute.

The buffer shall use a first-in-first-out eviction strategy when the storage limit is reached.

REQ-FUNC-005 Forwarding to Cloud

When connectivity is restored, the gateway shall forward buffered telemetry in chronological order.

The gateway shall not forward duplicate telemetry records. Each telemetry record shall include a stable event ID.

REQ-FUNC-006 Device Health

The gateway shall compute a health status for each registered device:

- healthy;
- degraded;
- disconnected;
- configuration_error.

Health status shall be updated within 30 seconds after a communication failure is detected.

REQ-FUNC-007 Alerting

The gateway shall generate an alert when:

- a device is disconnected for more than 2 minutes;
- the local buffer exceeds 80 percent capacity;
- cloud forwarding fails for more than 5 consecutive retry attempts;
- certificate expiration is less than 14 days away.

REQ-FUNC-008 Configuration Versioning

Every configuration update shall create a new immutable configuration version.

The user shall be able to compare the current configuration with the previous version.

REQ-FUNC-009 Rollback

The gateway shall allow a site administrator to roll back to the previous valid configuration version.

Rollback shall not delete the failed configuration version from the audit history.

REQ-FUNC-010 Diagnostic Bundle

The gateway shall generate a diagnostic bundle containing:

- system information;
- active configuration version;
- recent service logs;
- protocol adapter status;
- buffer statistics;
- last 100 audit events.

Sensitive secrets, passwords, private keys, and API tokens shall be redacted.

5. Non-Functional Requirements

REQ-NFR-001 Startup Time

After power-on, the gateway shall be ready to collect telemetry within 90 seconds.

REQ-NFR-002 Availability

The gateway software shall support continuous operation for at least 30 days without manual restart under nominal load.

REQ-NFR-003 Performance

The gateway shall process at least 2,000 measurements per minute with CPU usage below 70 percent on the reference hardware.

REQ-NFR-004 Time Synchronization

The gateway shall synchronize time using NTP. If NTP is unavailable, the gateway shall flag timestamps as unsynchronized.

REQ-NFR-005 Data Integrity

Telemetry records shall preserve:

- original timestamp;
- ingestion timestamp;
- device ID;
- signal ID;
- value;
- quality flag;
- event ID.

REQ-NFR-006 Security

All remote API access shall require authentication. Administrative actions shall require role-based authorization.

REQ-NFR-007 Auditability

The system shall retain audit events for at least 180 days or until exported by the site administrator.

6. Constraints

The first production release targets an ARM-based industrial gateway with:

- 4 CPU cores;
- 8 GB RAM;
- 64 GB local storage;
- Linux-based operating system;
- dual Ethernet interfaces.

The system shall not depend on permanent internet connectivity for local telemetry collection.

7. Out of Scope for Version 1

The following items are out of scope:

- machine learning inference at the edge;
- direct PLC firmware updates;
- high-frequency vibration streaming above 10 kHz;
- multi-site orchestration;
- mobile application.

8. Open Questions

1. Should the buffer retention target increase from 24 hours to 72 hours for high-priority sites?
2. Should OPC UA certificate rotation be managed automatically?
3. Should alert acknowledgements be synchronized to the cloud when connectivity returns?